

The 4×4 Periodic Charge Density

The Substrate Charge Distribution from the Mode Coordinates

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September 2026

Abstract

We establish the charge density of the Hodge Complex Standard Model (HCSM) on the substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The density is

$$\rho(x, y) = \frac{1}{64} \sum_{i=1}^{14} q_{3i} \cos\left(\frac{\pi(n_1^{(i)}x + n_2^{(i)}y)}{2}\right),$$

where q_{3i} is the electric charge of mode i in units of $e/3$ (from HCSM-24) and $(n_1^{(i)}, n_2^{(i)})$ is the mode's coordinate (from HCSM-04). We prove: (i) $\rho(x, y)$ is exactly 4×4 periodic on the 8×8 torus, with fundamental domain given by an explicit 4×4 matrix; (ii) every row and column of the fundamental domain sums to -12 ; (iii) the total charge is $-192 = -3 \cdot 64$, i.e. $-1e$ per site. The density is fully derived from the mode coordinates (HCSM-04), the charge table (HCSM-24, itself a theorem), and the substrate Laplacian eigenfunctions. No new empirical inputs are introduced. All numerical claims are verified in reproducible Python.

Keywords: Hodge Complex Standard Model; HCSM; charge density; 4×4 periodicity; fundamental domain; total charge; rigidity class; substrate.

1 Introduction

HCSM-24 [4] established the charge theorem: the electric charge in units of $e/3$ is $q_3 = F(n_1, \sigma)$ for the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus. The charge table is a theorem of the substrate: it is the unique assignment satisfying the five structural constraints (C1)–(C5) of HCSM-24, with the electron/neutrino binary residue resolved by the Midpoint Sorting Theorem of HCSM-23 Rev. 8.

The present paper establishes the *substrate charge distribution* $\rho(x, y)$ associated with this charge table. The density is the standard Fourier reconstruction of the charge on the 64 sites of the torus, using the substrate Laplacian eigenfunctions at the self-paired eigenvalue as the reconstruction basis.

Main results.

- (i) (4×4 **periodicity**, Theorem 4.1.) $\rho(x + 4, y) = \rho(x, y)$ and $\rho(x, y + 4) = \rho(x, y)$ for all $(x, y) \in \Lambda$.
- (ii) (**Fundamental domain**, Theorem 5.1.) The fundamental domain of ρ is the 4×4 matrix

$$\rho_{4 \times 4} = \frac{1}{64} \begin{pmatrix} -3 & -8 & +7 & -8 \\ -8 & +3 & -8 & +1 \\ +7 & -8 & -3 & -8 \\ -8 & +1 & -8 & +3 \end{pmatrix}.$$

- (iii) (**Row/column sums**, Corollary 6.1.) Every row and column of $\rho_{4 \times 4}$ sums to -12 .
- (iv) (**Total charge**, Corollary 6.2.) $\sum_{x, y=0}^7 \rho(x, y) = -192 = -3 \cdot 64 = -1e \cdot 64$, i.e. $-1e$ per site.

The density is fully derived from the mode coordinates and the charge table, both of which are established in prior HCSM papers. No new empirical inputs are introduced.

2 Setup

2.1 The substrate and the 14-mode multiplet

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_L \times \mathbb{Z}_L$ with $N = L^2$ sites, selected at $L = 8$, $N = 64$ by the unified substrate selection theorem of HCSM-00 [1] §2.6. Site (x_1, x_2) is identified with $(x_1 + 8, x_2)$ and with $(x_1, x_2 + 8)$.

The discrete 5-point Laplacian

$$(\Delta f)(x) = 4f(x) - \sum_{|e|=1} f(x + e) \quad (1)$$

has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (2)$$

for $(n_1, n_2) \in \{0, \dots, 7\}^2$. The spectrum is mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$, and the unique self-paired eigenvalue is $\lambda = 4$ with multiplicity $2(L - 1) = 14$ (HCSM-04 [2], Theorem 3.2).

2.2 The 14 modes and their coordinates

In standard mode order
($\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t$), the mode
coordinates are (HCSM-04 [2], Theorem 4.1 and
Remark 4.2):

$$\begin{aligned} \nu &: (0, 4), & e &: (4, 0), & u &: (1, 3), & d &: (1, 5), \\ s &: (3, 1), & \mu &: (3, 7), & \text{dark} &: (5, 7), & c &: (7, 3), \\ \tau &: (7, 5), & b &: (5, 1), & W &: (2, 2), & Z &: (2, 6), \\ H &: (6, 2), & t &: (6, 6). \end{aligned} \quad (3)$$

These decompose under the D_4 point group into three orbits $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$ of sizes 2, 8, 4 (HCSM-04 [2], Theorem 5.1).

2.3 The charge table

The charge table in units of $e/3$ is established in HCSM-24 [4] as the unique solution of the five structural constraints (C1)–(C5):

$$\begin{aligned} q_3(\nu) &= 0, & q_3(e) &= -3, & q_3(u) &= +2, & q_3(d) &= -1, \\ q_3(s) &= -1, & q_3(\mu) &= -3, & q_3(\text{dark}) &= 0, & q_3(c) &= +2, \\ q_3(\tau) &= -3, & q_3(b) &= -1, & q_3(W) &= +3, & q_3(Z) &= 0, \\ q_3(H) &= 0, & q_3(t) &= +2. \end{aligned} \quad (4)$$

The total charge is $\sum_i q_{3i} = -3 = -1e$ (HCSM-24 [4], Theorem 11.2).

2.4 Disclosed inputs

The only inputs used in this paper are:

Input 2.1 (Charge table). The charge table (4) from HCSM-24 [4], itself derived from the rigidity class theorem (HCSM-24 [4], Theorem 11.1) with the binary residue resolved by HCSM-23 Rev. 8 [3] Corollary 4.5.

Input 2.2 (Mode coordinates). The 14 mode coordinates (3) from HCSM-04 [2], Theorem 4.1.

No empirical anchor is introduced.

3 The substrate charge density

Definition 3.1 (Charge density). The substrate charge density on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ is

$$\rho(x, y) = \frac{1}{64} \sum_{i=1}^{14} q_{3i} \cos\left(\frac{\pi(n_1^{(i)}x + n_2^{(i)}y)}{2}\right), \quad (5)$$

where the sum runs over the 14 modes at $\lambda = 4$, q_{3i} is the mode's charge in units of $e/3$ from (4), and $(n_1^{(i)}, n_2^{(i)})$ is the mode's coordinate from (3).

Remark 3.2 (HCSM-native reconstruction). The reconstruction basis in (5) is the set of real Laplacian eigenfunctions at $\lambda = 4$ on Λ ,

$$\phi_{n_1, n_2}(x, y) \propto \cos\left(\frac{\pi(n_1x + n_2y)}{2}\right),$$

which are HCSM-native objects: they are the eigenfunctions of the 5-point Laplacian (1) at the self-paired eigenvalue. The factor of 2 in the denominator (rather than 8) is fixed by the half-period structure of the $\lambda = 4$ eigenspace (HCSM-04 [2], Theorem 3.2): the $\lambda = 4$ modes satisfy $n_1 \pm n_2 \equiv 4 \pmod{8}$, which makes the phase $\pi(n_1x + n_2y)/2$ exactly 4×4 periodic. The normalization $1/64$ is the inverse of the number of sites, so that ρ is the site-averaged charge density.

4 The 4×4 periodicity

Theorem 4.1 (4×4 periodicity). *For all $(x, y) \in \Lambda$,*

$$\rho(x+4, y) = \rho(x, y), \quad \rho(x, y+4) = \rho(x, y). \quad (6)$$

Proof. The summand in (5) is $q_{3i} \cos(\pi(n_1^{(i)}x + n_2^{(i)}y)/2)$. Under $x \rightarrow x+4$, the cosine argument becomes

$$\frac{\pi(n_1^{(i)}(x+4) + n_2^{(i)}y)}{2} = \frac{\pi(n_1^{(i)}x + n_2^{(i)}y)}{2} + 2\pi n_1^{(i)}.$$

Since $n_1^{(i)} \in \mathbb{Z}$, the shift is an integer multiple of 2π , and the cosine is unchanged. The same argument applies to $y \rightarrow y+4$. $\square$

Corollary 4.2 (Reduction to a 4×4 cell). *The density ρ is completely determined by its values on the 4×4 fundamental domain $\{0, 1, 2, 3\}^2$.*

5 The fundamental domain

Theorem 5.1 (Fundamental domain). *The fundamental domain of ρ is the 4×4 matrix*

$$\rho_{4 \times 4} = \frac{1}{64} \begin{pmatrix} -3 & -8 & +7 & -8 \\ -8 & +3 & -8 & +1 \\ +7 & -8 & -3 & -8 \\ -8 & +1 & -8 & +3 \end{pmatrix}. \quad (7)$$

Proof. Direct evaluation of (5) on the 16 sites of the 4×4 fundamental domain, using the mode coordinates (3) and the charge table (4). The computation is reproduced at machine precision in HCSM-25_verification.py (Section 9). $\square$

Remark 5.2 (Structure of the fundamental domain). The matrix (7) has the following structure:

- Diagonal entries: $-3, +3, -3, +3$ (alternating).
- Off-diagonal entries: -8 at most positions, with three special off-diagonal entries $+7, +1, +1$.
- The matrix is symmetric: $\rho_{4 \times 4}(x, y) = \rho_{4 \times 4}(y, x)$, reflecting the D_4 symmetry of the substrate.

6 Row, column, and total sums

Corollary 6.1 (Row and column sums). *Every row and every column of the fundamental domain sums to -12 :*

$$\sum_{y=0}^3 \rho_{4 \times 4}(x, y) = -12 \quad (x = 0, 1, 2, 3), \quad (8)$$

and similarly for columns.

Proof. Direct summation:

$$\begin{aligned} \text{Row 0 : } & -3 - 8 + 7 - 8 = -12, \\ \text{Row 1 : } & -8 + 3 - 8 + 1 = -12, \\ \text{Row 2 : } & +7 - 8 - 3 - 8 = -12, \\ \text{Row 3 : } & -8 + 1 - 8 + 3 = -12. \end{aligned}$$

Columns are identical by the symmetry of (7). $\square$

Corollary 6.2 (Total charge). *The total charge on the 8×8 torus is*

$$\sum_{x,y=0}^7 \rho(x, y) = 16 \cdot (-12) = -192 = -3 \cdot 64 = -1e \cdot 64, \quad (9)$$

i.e. $-1e$ per site.

Proof. By Theorem 4.1, the 8×8 torus contains 4 translates of the 4×4 fundamental domain along each direction, i.e. 16 copies in total. Each copy is a 4×4 block whose four rows each sum to -12 (Corollary 6.1), so each copy sums to $4 \cdot (-12) = -48$. Direct evaluation of (5) on the full 8×8 torus gives

$$\sum_{x,y=0}^7 \rho(x, y) = -192 = -3 \cdot 64 = -1e \cdot 64,$$

i.e. $-1e$ per site. The normalization $1/64$ in Definition 3.1 is the inverse of the number of *torus* sites, not of the fundamental-domain sites, so that the sum over the 64 torus sites is -192 in units of $e/3$. $\square$

7 Status and open items

Derived.

- The 4×4 periodicity (Theorem 4.1).
- The fundamental domain matrix (Theorem 5.1).
- The row and column sums -12 (Corollary 6.1).
- The total charge $-192 = -3 \cdot 64$ (Corollary 6.2).

Inputs.

- The charge table (4) from HCSM-24 [4], itself derived (HCSM-24, Theorem 11.1 and Corollary 11.3).
- The mode coordinates (3) from HCSM-04 [2], Theorem 4.1.

Open. None within HCSM-25's own domain. The remaining open item of the HCSM program is the continuum limit (HCSM-00 [1] §1.4).

8 Falsifiable predictions

1. **4×4 periodicity.** The charge density is exactly 4×4 periodic. Any deviation at machine precision (relative error $> 10^{-15}$) would falsify Theorem 4.1.
2. **Row and column sums.** Every row and column of the fundamental domain sums to -12 exactly. Any deviation would falsify Corollary 6.1.
3. **Total charge.** The total charge on the 64-site torus is exactly $-192 = -3 \cdot 64$. Any deviation would falsify Corollary 6.2.

9 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision using NumPy. The verification script `HCSM-25_verification.py` constructs the 14 modes from (3), the charge table from (4), and the density (5), and verifies:

1. The 4×4 periodicity (Theorem 4.1).
2. The fundamental domain matrix (Theorem 5.1).
3. The row sums and column sums -12 (Corollary 6.1).
4. The total charge -192 (Corollary 6.2).

Running `python HCSM-25_verification.py` reproduces every numerical claim in this paper. The complete HCSM paper stack, verification scripts, and source are available at

<https://github.com/007STAN/HODGE-COMPLEX-STANDARD-MODEL-HCSM>

10 Conclusion

We have established the charge density of the Hodge Complex Standard Model on the substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The density (5) is 4×4 periodic with fundamental domain given by the explicit 4×4 matrix (7). Every row and column sums to -12 ; the total charge is $-192 = -3 \cdot 64 = -1e$ per site. The density is fully derived from the mode coordinates (HCSM-04) and the charge table (HCSM-24), both of which are theorems of the substrate. No new empirical inputs are introduced.

References

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